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EARLY DROPOUT FROM UNIVERSITY MATHEMATICS: THE ROLE OF STUDENTS' ATTITUDES TOWARDS MATHEMATICS

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High dropout rates in mathematics during the first year at university underline students' difficulties in the transition from school to university mathematics. In this contribution, we present a quantitative study based on the three-dimensional model for attitude towards mathematics. Using questionnaires, we analyse differences concerning attitudes towards mathematics between dropped out students and students who continued their studies of mathematics. Our results show that dropped out students are less interested in university mathematics and report a lower mathematical self-concept than those students who continued their studies. Moreover, dropped out students report a decline of their mathematical self-concept during the transition from school to university mathematics.

INTRODUCTION

Dropout is a major concern in university mathematics. In Germany nearly 80 % of all mathematics students drop out or change their subject (Dieter & Törner, 2012) – most of them during their first year at university, so called early dropout. These facts reveal students' difficulties during the transition from school to university mathematics.

One obstacle during the transition are the major differences between mathematics at school and at university. These differences have been extensively discussed in the literature – for a detailed discussion see Ufer, Rach and Kosiol (2017). At school, new concepts are introduced with many examples aiming at an intuitive understanding. In contrast, new concepts at university are introduced via formal definitions. Whereas tasks in school mathematics are often focused on solving real-world problems and schematic calculations, typical tasks at university involve proofing (cf. Ufer et al., 2017). These tasks usually are not directly connected to the real world and cannot be solved by schematic calculations. In Germany, freshmen traditionally attend the courses Real Analysis and Linear Algebra which are focussed on formal definitions and deductive proofs (Halverscheid & Pustelnik, 2013). Both courses are usually accompanied by weekly tutorials and homework tasks with a strong focus on proofs.

According to theories of person-environment-fit (e.g. Swanson & Fouad, 1999) a sufficient fit between the characteristics of the students (e.g. attitudes, prior

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knowledge, learning behaviour) and the characteristics of the university (e.g. contents, learning environment) is necessary for a successful transition. Due to the already mentioned differences between mathematics at school and at university, this fit does not seem to be self-evident. However, a sufficient fit between the characteristics of the students and the characteristics of the university leads to satisfaction and appropriate achievements of the students (Swanson & Fouad, 1999) while an insufficient fit increases the risk for dropout. According to Haak (2017), an insufficient fit leads to a personal crisis. This crisis is frequently mentioned in the mathematics education literature (e.g. Di Martino & Gregorio, 2019). Haak (2017) proposes two ways to overcome this crisis: Students can either adapt their personal characteristics (e.g. their attitudes or their learning behaviour) or they can decide to drop out.

While early studies in the field of transition to university mathematics had mainly a cognitive orientation (Artigue, 2016), recent research pays attention to affect and the role of attitudes during the transition as well (e.g. Rach & Heinze, 2017; Di Martino & Gregorio, 2019). In this contribution, we focus on the role of students' attitudes towards mathematics for early dropout from university mathematics.

ATTITUDES TOWARDS MATHEMATICS

The question how to conceptualise attitudes and mathematics related affect has been frequently discussed in the mathematics education literature (Di Martino & Zan, 2011). Di Martino and Zan (2011) have proposed a Three-dimensional Model for Attitude (TMA) towards mathematics (comprising the dimensions *emotional disposition, vision of mathematics* and *perceived competence*), which is based on an analysis of school students' narratives about their attitudes to and experiences with mathematics. Following the TMA, we understand attitudes towards mathematics as an interplay between interest in mathematics (*emotional disposition*), beliefs concerning the nature of mathematics (*vision of mathematics*) and mathematical self-concept (*perceived competence*) (cf. Di Martino & Gregorio, 2019). The TMA has already been used in qualitative studies focussing on students' experiences and difficulties during the transition to university mathematics (e.g. Di Martino & Gregorio, 2019).

Interest in Mathematics

Individual interest is considered to be a rather stable relationship between an (abstract) object and a person, comprising of emotional (e.g feeling of joy while engaging with the object of interest) and value related (e.g. personal esteem of the object of interest) components (Krapp, 2007). Since people are motivated to engage with the objects of interest, interest is considered to play a crucial role for successful learning processes.

Studies concerning the role of interest during the transition from school to university mathematics have reported contradictory results (cf. Ufer et al.,

2017). Ufer et al. (2017) have argued, that a clear distinction between interest in school mathematics and interest in university mathematics is necessary, when dealing with interest during the transition. Otherwise it is not clear whether students have school or university mathematics in mind, while answering items measuring interest in mathematics. Kosiol, Rach and Ufer (2019) found that interest in university mathematics goes hand in hand with more satisfaction during the first term at university, while interest in school mathematics is connected with less satisfaction. We follow the argumentation of Ufer et al. (2017) and differentiate between interest in school mathematics and interest in university mathematics in this contribution.

Beliefs Concerning the Nature of Mathematics

Philipp (2007, p. 259) describes beliefs as "propositions about the world that are thought to be true". Traditionally we distinguish between rather static and dynamic beliefs concerning the nature of mathematics (Grigutsch & Törner, 1998). Static beliefs are characterized by the view that mathematics is a static summary of different (unconnected) procedures, rules and formula. In contrast, dynamic beliefs highlight that mathematics is a creative process and field of research with applications in other domains and everyday life.

Regardless that university teachers hold static as well as dynamic beliefs (Grigutsch & Törner, 1998), dynamic beliefs seem to be more beneficial than static ones during the transition. Dynamic beliefs correlate positive with interest in mathematics during the first year at university (Liebendörfer & Schukajlow, 2017). Moreover, students with rather dynamic beliefs are more successful in exams than students with rather static beliefs (Crawford, Gordon, Nicholas & Prosser, 1994).

Mathematical Self-Concept

Bong and Skaalvik (2003) describe self-concept as a person's perception about herself or himself with emphasis on the own skills and abilities. The self-concept is influenced by prior experiences especially mastery experiences and the feeling of competence and success in a particular domain (Bong & Skaalvik, 2003).

Di Martino and Gregorio (2019) found that most mathematics students start their studies with a high mathematical self-concept but report decreasing self-concept during the transition due to experiences of failure. Rach and Heinze (2017) found that students' mathematical self-concept predicts their exam attendance in the first term at university. Students who do not attend their exams – according to Baars and Arnold (2014) an useful indicator for dropout – report a lower mathematical self-concept than those students who attend the exams.

RESEARCH QUESTIONS AND METHODS

In this contribution, we want to clarify the role of students' attitudes towards mathematics for early dropout during the first year at university. Since the character of mathematics changes during the transition from school to university, changes in students' attitudes during this transition phase are likely. Furthermore, an adaption of attitudes is one possibility to overcome the crisis that occurs if students' attitudes do not fit to the characteristics of university mathematics (Haak, 2017). Therefore, we consider students' attitudes at the beginning of the first term and during the first term at university. This leads to the following questions and hypotheses:

1) Do students who dropped out from mathematics and students who continued with their studies already differ concerning their attitudes towards mathematics at the beginning of the first term?
2) Do students who dropped out from mathematics and students who continued with their studies differ concerning their attitudes towards mathematics in the middle of the first term?

Overall, we expect the differences between the two groups of students during the first term to be larger but in the same direction as the differences at the beginning of the term. In detail, we phrase the following three hypotheses:

Since interest in university mathematics goes hand in hand with more satisfaction (Kosiol et al., 2019), we expect that students who dropped out report less interest in university mathematics than those students who continued their studies of mathematics (H1). We have no special hypothesis concerning the interest in school mathematics.

With regard to beliefs concerning the nature of mathematics, dynamic beliefs seem to be more beneficial for a successful transition than static beliefs (Crawford et al., 1994). That is why we expect that dropped out students tend to agree more to static beliefs and less to dynamic beliefs than those students who continued their studies (H2).

Based on the result that students with low mathematical self-concept often do not attend their exams (Rach & Heinze, 2017) – which is an indicator for dropout – we believe that dropped out students will report a lower mathematical self-concept than those students who continued their studies of mathematics (H3).

In order to answer the research questions, two questionnaires – one at the beginning of the first term (at the end of the second week, T1) and one in the middle of the first term (at the end of the ninth week, T2) – were used. At the end of the first year we checked whether students continued their studies or dropped out. The questionnaires have been handed out in the Real Analysis and the Linear Algebra lectures (which German freshmen usually attend during their

first term) at a large public German university. The instruments used in the questionnaires can be found in table 1.

Variable	Source	#	α(T1/T2)	Example
Interest School Mathematics	Ufer et al., 2017	5	0,80/ 0,80	In school, mathematics was very important for me.
Interest University Mathematics		5	0,88/0,87	The kind of mathematics that is done at university is fun for me.
Beliefs: static	Laschke & Blömeke, 2013	6	0,52/ 0,64	Mathematics means learning, remembering and applying.
Beliefs: dynamic		6	0,71/ 0,76	Mathematics involves creativity and new ideas.
Mathematical Self-Concept	Kauper et al., 2012	4	0,84/ 0,82	I am very good in my study subject mathematics.

Table 1: Instruments used in the questionnaire with number of items (#) and reliability (cronbachs α)

All items had to be answered on a five-point likert scale (1= totally disagree; 5=totally agree). 271 freshmen (mathematics majors and mathematics pre-service teachers) voluntarily filled out the first questionnaire. 222 freshmen participated in the second survey. All used scales had at least satisfying reliability, except the static beliefs scale, which has therefore been excluded from the further data analysis.

RESULTS

To answer the research questions and to check the hypotheses, multivariate analyses of variance (MANOVA) were used – to avoid the cumulation of the α-error compared with single *t*-tests. In the following, we first describe the results concerning students' attitudes at the beginning of the first term (T1), before discussing the results concerning students' attitudes during the term (T2).

Attitudes at the Beginning of the first Term (T1)

A large group of students who dropped out during their first year did not attend the lectures at the middle of the first term (T2) anymore. Therefore, we compare three groups of students: students who continued their studies (no dropout), students who dropped out prior to T2 (very early dropout) and students who dropped out during the first year but after T2 (early dropout). Table 2 shows the results concerning the differences between these groups:

Variable	Very Early Dropout N=59		Early Dropout N=40		No Dropout N=172		η^2
	M	*SD*	*M*	*SD*	*M*	*SD*	
Interest School Mathematics	3,41	0,65	3,45	0,63	3,62	0,69	0,02
Interest University Mathematics	2,55	0,79	2,86	0,78	3,17	0,83	0,09***
Beliefs: dynamic	3,50	0,54	3,60	0,53	3,74	0,59	0,03*
Mathematical Self-Concept	2,41	0,62	2,76	0,58	2,97	0,70	0,10***

Table 2: Means, standard deviations and results of the MANOVA concerning the attitudes towards mathematics at T1; N=271; *p<0.05; ***p<0.001

As expected, dropped out students report less interest in university mathematics (H1), less dynamic beliefs (H2) and a lower self-concept (H3). No significant differences can be found with regard to interest in school mathematics. Due to the fact, that the static beliefs have been excluded from the analysis, H2 can only be confirmed partially.

Post-Hoc-tests (with Bonferroni correction) show that mainly the differences between the very early dropped out students and the students who continued their studies are significant. Furthermore, the very early dropped out students report significantly lower self-concept than the early dropped out ones. However, early dropped out students do not differ significantly in their attitudes from students who continued their studies.

Attitudes during the First Term (T2)

Since the very early dropped out students did not attend the lectures at T2 anymore, differences concerning the attitudes during the first term can only be compared between early dropped out students and those students who continued their studies. Table 3 shows these differences. As expected dropped out students report significantly less interest in university mathematics than the students who continued their studies (H1). We found no significant differences with regard to interest in school mathematics and dynamic beliefs. However, early dropped out students report significantly less mathematical self-concept than the students that continued their studies (H3).

Variable	Early Dropout N=56		No Dropout N=166		η^2
	M	*SD*	*M*	*SD*	
Interest School Mathematics	3,62	0,72	3,70	0,72	0,00
Interest University Mathematics	2,97	0,79	3,27	0,81	0,03*
Beliefs: dynamic	3,40	0,60	3,42	0,64	0,00
Mathematical Self-Concept	2,51	0,63	2,95	0,66	0,06***

Table 3: Means, standard deviations and results of the MANOVA concerning the attitudes towards mathematics at T2; N=222; *p<0.05; ***p<0.001

This is remarkable because at the beginning of their studies, these two groups of students did not differ significantly concerning their mathematical self-concept. A closer look at the means at T1 and T2 reveal that the mathematical self-concept of those students who continued their studies remains nearly constant while the early dropped out students report a clear decline of their self-concept.

DISCUSSION

Our results indicate that dropped out students and students who continued their studies of mathematics differ mainly concerning their interest in university mathematics and their self-concept. The differences found depend on the time of measurement.

At the beginning of the first term, only the very early dropped out students report less interest in mathematics, less agreement to dynamic beliefs and a lower mathematical self-concept than the students who continued their studies. It seems that the very early dropped out students start their studies of mathematics with unfavourable attitudes that do not fit to university mathematics (in the sense of person-environment-fit). It seems that these students do not try to adapt their attitudes (as proposed by Haak (2017)) and therefore drop out very fast. It remains the question, whether some kind of supporting program would be beneficial for this group or if the very early dropout has to be understood as a fast correction of a wrong study choice. In this case it would be helpful to inform future students better about mathematics at university to enable them to make deliberate and appropriate study choices. Especially information about major differences between mathematics at school and at university should be given – preferably already during secondary school.

The early dropped out students did not differ from those students who continued their studies concerning their attitudes towards mathematics at the beginning of the first term. However, in the middle of the first term, the early dropped out

students report significantly less interest in university mathematics and a lower mathematical self-concept than the students who continued their studies. While the self-concept of the students who continued their studies remained nearly constant during the transition, the self-concept of the early dropped out students decreased. This is in line with the findings of qualitative studies like the one of Di Martino and Gregorio (2019) who found that the mathematical self-concept of many students – even those that continued their studies – decreases during the transition from school to university due to the unexpected experiences of failure in mathematics. Many experiences of failure during the transition are connected to students' problems with the weekly homework tasks (Liebendörfer & Hochmuth, 2017). Therefore, the design of these tasks could be reconsidered. Tasks that are challenging but offer experiences of success might help strengthening students' mathematical self-concept.

All in all, we found clear evidence for differences concerning the attitudes towards mathematics between students who dropped out and those who continued their studies. However, our study has some limitations. We only collected data at one university, thus our results might only reflect the local situation. The questionnaires relied on self-reports that can be biased. In addition, questionnaires were filled out during lectures. Students who do not regularly attend the lectures were not captured in our study. Our ongoing research will now focus on supporting measures that foster students' mathematical self-concept.

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